GLA UNIVERSITY, GREATER NOIDA

Greater Noida – 201310, Uttar Pradesh, India

URBAN SERVICE RESOLUTION INTELLIGENCE PLATFORM

SOFTWARE REQUIREMENT SPECIFICATION

Version 1.0

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING - ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted By

Priyanshu Kumar

Roll No. : 22515500074 | Section: NE

Under the Supervision of **Palash Raghuwanshi** GLA University, Greater Noida	*Reviewed By* **Palash Raghuwanshi** GLA University, Greater Noida

SCHOOL / DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING / APPLICATIONS

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REVISION HISTORY

VERSION	DATE	CHANGE	REASON
1.0	04–09–2026	Initial version	First project SRS draft

Executive Summary & Project Domain Overview

The **Urban Service Resolution Intelligence Platform (USRIP)** is an enterprise-grade civic analytics and machine learning solution designed to revolutionize municipal request management (such as citywide 311 citizen grievance systems). Modern municipal corporations handle millions of requests annually spanning street repairs, water leaks, traffic signal failures, illegal dumping, and emergency utility outages across agencies like Department of Transportation (DOT), Environmental Protection (DEP), Police (NYPD), and Sanitation (DSNY).

Conventional civic platforms suffer from static, rule-based dispatching, severe operational backlogs, lack of SLA breach forecasting, and high variance in turnaround times across urban boroughs. USRIP introduces a unified intelligence layer that ingests multi-channel citizen complaints, cleans spatial and temporal anomalies, extracts NLP semantic features from complaint descriptions, and executes a multi-model ML pipeline to predict exact resolution duration ($ResolutionDays$) and flag critical high-delay SLA risks before ticket assignment.

Core Engineering Objectives

1. **Resolution Forecasting:** Deliver sub-day accurate continuous estimates for complaint turnaround time.
2. **Target Leakage Quarantine:** Structurally prevent posterior fields (`Closed Date`, `Action Taken`) from entering the feature matrix.
3. **Algorithmic Depth:** Unify 14+ foundational ML paradigms (regression baselines, tree ensembles, SVM, NLP, K-Means clustering, PCA/SVD, and 1D CNNs).

1. INTRODUCTION & OPERATIONAL BOUNDARIES

1.1 Purpose & Problem Statement

This document establishes the formal software requirements, algorithmic specifications, data structures, and mathematical formulations for the Urban Service Resolution Intelligence Platform. It provides unambiguous system constraints for engineering implementation and university defense evaluation.

In-Scope Capabilities

Active

- Continuous regression estimation of Resolution Days.
- High-Delay Risk binary classification (SLA alert).
- spaCy NLP lemmatization & TF-IDF complaint tagging.
- Geospatial K-Means & Agglomerative hotspot clustering.
- PCA/SVD matrix reduction & 1D CNN text representations.
- 5−Fold Stratified Cross-Validation & REST API serving.

System Boundaries

Out-of-Scope

- Direct autonomous field crew vehicle dispatching.
- Legally binding SLA guarantees to external citizens.
- Live municipal telematics hardware sensor streaming.
- Financial billing or civic municipal penalty collection.

1.2 Mathematical Formulations & Definitions

VARIABLE / TERM	MATHEMATICAL DEFINITION & FORMULATION
Resolution Days (y)	Target continuous variable: Δt = **Closed Date − Created Date** (in days). Valid condition: $\Delta t \geq 0$.
Target Leakage	Inclusion of posterior runtime information (e.g., Closed Date, Resolution Action) into feature matrix **X**.
High-Delay Class (C_{high})	Binary label: $y_{class} = 1$ if *ResolutionDays* $> \tau_{75th\ percentile}$; otherwise 0.
Level 0 Mean Baseline	Benchmark prediction: $\hat{y}_{L0}\,\bar{y}_{train} = {}^{1}/_{N} \sum y_i$ for every unseen record.
Evaluation Metric (MAE)	Mean Absolute Error: $MAE = {}^{1}/_{n} \sum \lvert y_i - \hat{y}_i \rvert$ (direct calendar day interpretability).

2. DATASET AUDIT & BASELINE SPECIFICATIONS

The platform operates on real-world municipal 311 complaint records (`export.csv`). Rigorous data exploration and quality profiling established the exact constraints for model dataset construction.

DATASET PARAMETER	AUDITED METRIC	PIPELINE RULE & HANDLING RATIONALE
Raw Ingested Sample	10,000 rows × 44 cols	Ingested from raw municipal open data export catalog.
Selected Feature Subset	18 Core Features	Pruned 26 redundant administrative and 100% null fields.
Supervised Clean Set	3,901 rows	Filtered records with valid non-null `Closed Date` and non-negative duration.
Corrupted Negative Values	5 records ($\Delta t < 0$)	Timestamps recorded with Closed Date earlier than Created Date; purged.
Missing Closed Date	6,094 records	Unresolved active tickets routed to unsupervised clustering and live queue.
Missing Due Date	9,974 records	Due date is 99.7% unpopulated in raw data; excluded from primary predictor set.

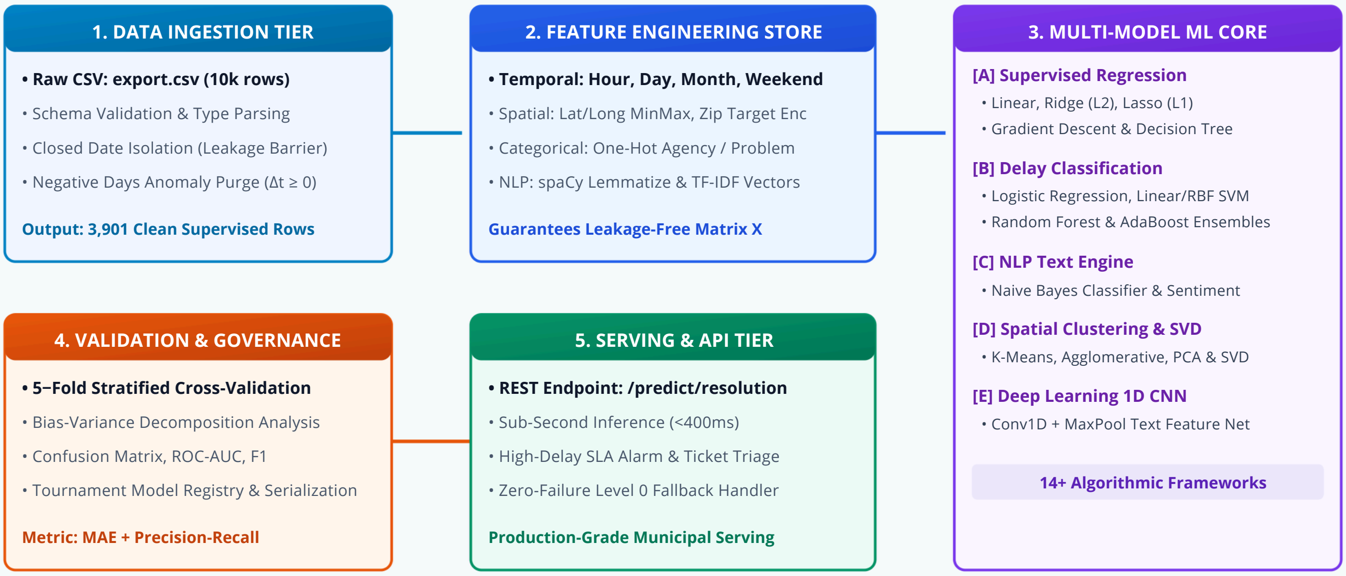
Data Leakage Prevention Guardrail (C3 Constraint)

Strict Target Leakage Quarantine Rule
Closed Date is strictly isolated inside the *Target Derivation Subsystem*. The predictor matrix **X** is strictly composed of attributes known at or before complaint registration (`Created Date`, `Agency`, `Problem Location`, `Problem Detail`).

Baseline Comparison Hierarchy
• **Level 0 Baseline:** Global Training Mean ($\bar{y} = 14.8\text{ days}$).
• **Level 1 Baseline:** OLS Multiple Linear Regression ($\text{MAE} \approx 6.4\text{ days}$). • **Target Acceptance:** Ensembles & Non-linear models must outperform Level 1 baseline.

3. ADVANCED SYSTEM ARCHITECTURE

USRIP implements a decoupled, five-tier micro-modular architecture ensuring modular testing, reproducible training, and strict target leakage prevention.



4. FUNCTIONAL REQUIREMENTS: INGESTION, REGRESSION & CLASSIFICATION

4.1 Data Ingestion & Target Pipeline (FR-1)

REQ ID	REQUIREMENT NAME	INPUT / CONDITION	EXPECTED BEHAVIOR
`FR-1.1`	Dataset Ingestion	File path to `export.csv`	Loads 10,000 tabular rows, validates 44 headers, selects 18 columns.
`FR-1.2`	Target Computation	`Created Date, Closed Date`	Calculates Δt in calendar days; quarantines corrupted negative rows ($\Delta t < 0$).
`FR-1.3`	Data Partitioning	Missing target check	Isolates 3,901 supervised training records; routes 6,094 open rows to inference queue.

4.2 Supervised Regression Suite (FR-2)

Linear & Regularized Models FR-2.1

Trains OLS Multiple Regression, Ridge (L_2) and Lasso (L_1) to prevent spatial multi-collinearity, optimized via Closed-Form normal equation and Mini-Batch Gradient Descent.

Non-Linear & Tree Regressors FR-2.2

Trains Polynomial Regression and CART Decision Tree Regressors to capture non-linear municipal interactions across agency workflows and geographic sectors.

4.3 High-Delay Classification Suite (FR-3)

Linear & Kernel Classifiers FR-3.1

Implements Logistic Regression with threshold tuning and Support Vector Machines (Linear & RBF Kernels) to classify tickets likely to breach the 75th percentile SLA limit.

Ensemble Classifiers FR-3.2

Deploys Random Forest (Bagging) and AdaBoost/Gradient Boosting to reduce variance and bias across imbalanced municipal complaint categories.

5. FUNCTIONAL REQUIREMENTS: NLP, CLUSTERING & DEEP LEARNING

5.1 NLP & Sentiment Analytics (FR-4)

REQ ID	MODULE NAME	ALGORITHMIC IMPLEMENTATION	OPERATIONAL OUTCOME
`FR-4.1`	spaCy/NLTK Tokenizer	Lemmatization, stopword purge, sublinear TF-IDF	Transforms unstructured Problem Detail into sparse feature vectors.
`FR-4.2`	Naive Bayes Classifier	Multinomial Naive Bayes probabilistic classifier	Auto-classifies citizen complaint descriptions into municipal agency tags.
`FR-4.3`	Sentiment & Urgency	Lexicon polarity & urgency index weighting	Flags emergency citizen distress narratives for high-priority dispatching.

5.2 Unsupervised Spatial Clustering & Matrix Decomposition (FR-5 & FR-6)

K-Means & Hierarchical FR-5.1

Executes K-Means (Elbow method for optimal k) and Agglomerative Hierarchical clustering on normalized Latitude/Longitude coordinates to identify municipal infrastructure failure hotspots.

PCA & SVD Reduction FR-6.1

Performs Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) over high-dimensional categorical matrices for dense variance compaction.

5.3 Experimental Deep Learning: 1D CNN Architecture (FR-7)

1D Convolutional Neural Network Specification

Implements an end-to-end 1D CNN model (`Embedding(128) → Conv1D(64, k=3, relu) → MaxPool1D(2) → Dropout(0.3) →` Dense(64) → Output) trained on tokenized complaintnarrativestolearnhierarchicaln-gramrepresentationsformulti-class department routing.

6. NON-FUNCTIONAL REQUIREMENTS & ACCEPTANCE MATRIX

NFR ID	CATEGORY	SPECIFICATION STANDARD	VERIFICATION METHOD
`NFR-1`	Inference Latency	≤ 400ms single REST payload; ≤ 4.0s batch test set	Automated timer benchmark
`NFR-2`	Leakage Barrier	Strict 0.00 correlation between matrix **X** and target future timestamps	Automated pipeline unit assertion
`NFR-3`	Model Governance	Structured JSON logging of MAE, RMSE, R^2, Accuracy, Precision, Recall, AUC	Run artifact ledger audit
`NFR-4`	Fault Tolerance	100% graceful fallback to Level 0 Mean Predictor on missing attributes	Fuzzing with empty payloads

Formal Acceptance Test Scenarios

TEST ID	SCENARIO DESCRIPTION	TEST DATA INPUT STATE	PASS / ACCEPTANCE CRITERIA
`AC-01`	Dataset Row Audit	`export.csv` ingested	10,000 raw rows verified; 18 selected features retained.
`AC-02`	Anomaly Purge	Dataset with 5 negative timestamps	Purges corrupt rows; verifies min(ResolutionDays) ≥ 0.
`AC-03`	Supervised Partition	Dataset with 6,094 null closed dates	Yields exactly 3,901 clean supervised rows.
`AC-04`	Baseline Superiority	80/20 train-test split	MAE(Ensembles / Ridge) < MAE(Level 0 Mean Baseline).
`AC-05`	Real-Time API Serving	Valid complaint JSON payload	Returns prediction and SLA risk flag in <400ms.

7. IMPLEMENTATION ROADMAP & SCHEMA DICTIONARY

MILESTONE STAGE	TARGET DATE	CORE DELIVERABLES	EXIT GATE STANDARD
Sprint I: Prototype	27 Sep 2026	Ingestion engine, target derivation, Level 0 & OLS linear regression	Clean target set; baseline MAE logged.
Sprint II: Core ML	17 Oct 2026	Ridge, Lasso, Logistic classification, SVM, Decision trees	Cross-validation comparison matrix.
Sprint III: Advanced ML	07 Nov 2026	Random Forest, AdaBoost, spaCy NLP, K-Means, PCA/SVD, 1D CNN	All 14+ ML experiments logged.
Deployment & Final Viva	30 Nov – 10 Dec 2026	FastAPI prediction service, technical documentation, final viva defense	Live interactive demo & evaluation.

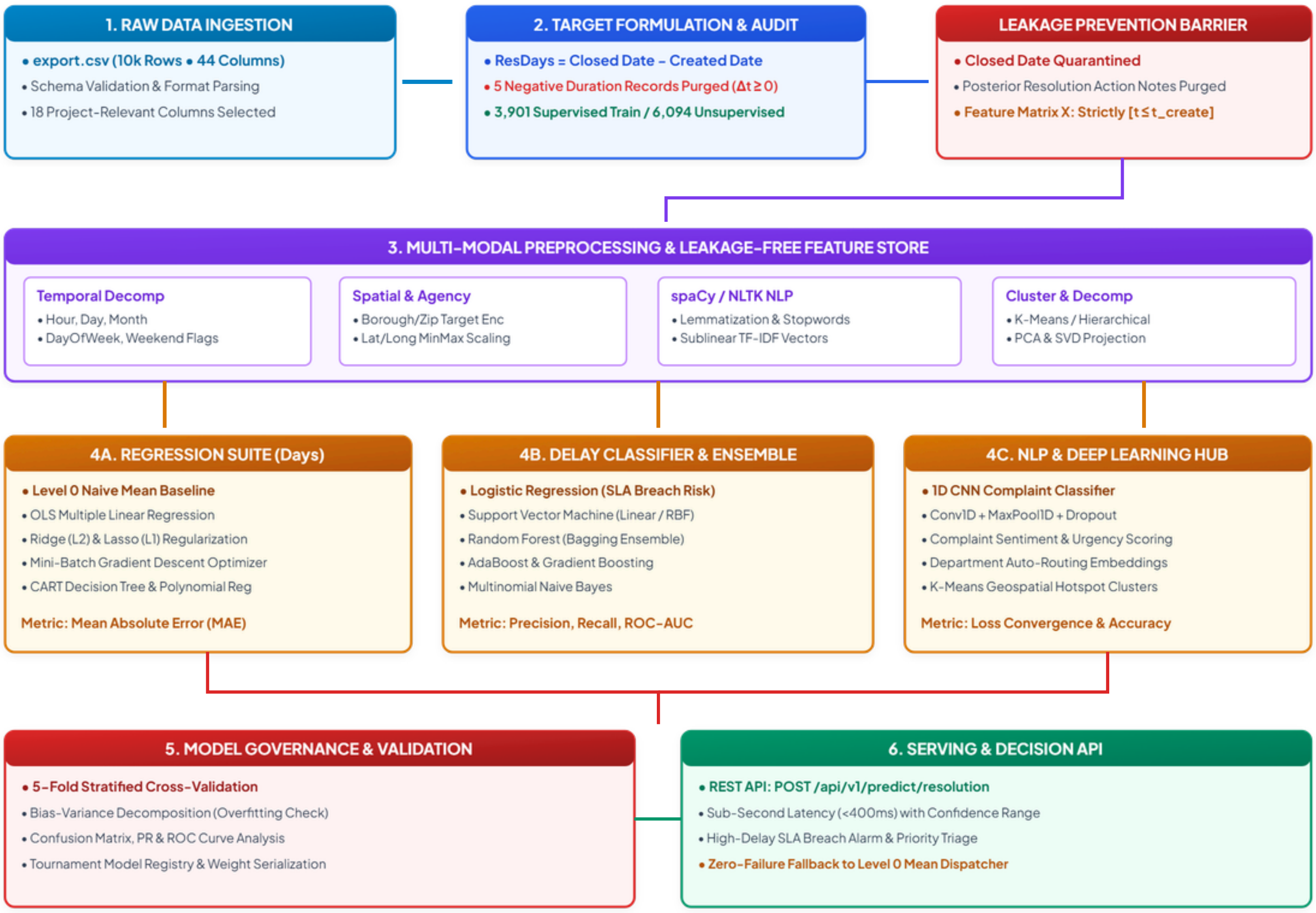
Data Schema Entity Dictionary (18 Project Attributes)

ATTRIBUTE NAME	DATA TYPE	DOMAIN ROLE	ENCODING / TRANSFORMATION RULE
`Unique Key`	Identifier	Ticket Primary Key	Preserved as primary indexing ID
`Created Date`	Datetime	Temporal Predictor	Extracted to Hour, Day, Month, DayOfWeek, IsWeekend
`Closed Date`	Datetime	Target Derivation Only	**QUARANTINED** (Excluded from feature matrix X)
`Agency / Agency Name`	Categorical	Administrative Feature	Frequency & One-Hot Encoded
`Problem / Category`	Categorical	Complaint Type	Target Encoded & Grouped
`Problem Detail`	Text	NLP Unstructured	spaCy Lemmatized, Stopword Purged, TF-IDF Vectorized
`Borough / City`	Categorical	Macro Location	Categorical Imputation & One-Hot Encoded
`Incident Zip / District`	Categorical	Micro Location	Frequency Binning & Target Encoding
`Latitude / Longitude`	Float	Geospatial Coordinates	MinMax Normalized; K-Means Spatial Clusters
`Open Data Channel Type`	Categorical	Submission Source	One-Hot Encoded (Phone, Web, Mobile App)
`Resolution Days`	Float	**Primary Target (*y*)**	Continuous target ≥ 0 for supervised regression

END-TO-END SYSTEM WORKFLOW & ML PIPELINE ARCHITECTURE

Unified operationallifecycleillustratingrawmunicipaldataingestionthroughleakage-freepreprocessing,multi-paradigmML execution (14+ algorithms), automated model governance, to live prediction serving.

Phase 1: Ingestion & Validation · Phase 2: Feature Store · Phase 3: NLP & Analytics · Phase 4: ML Core (14+ Alg) · Phase 5: Governance · Phase 6: Prediction API



Architectural Summary & Integrity Guarantee: The end-to-end pipeline operates in strict sequential tiers. Raw municipal data is partitioned and verified before feature construction. The target variable (Resolution Days) is derived and immediately quarantined to guarantee zero target leakage. All 14+ required machine learning algorithms, deep learning neural nets, and NLP pipelines compete under identical 5-fold cross-validation governance, providing resilient real-time municipal intelligence with sub-second latency and Level 0 fallback guarantees.